

LINUX PROGRAMMING: ASSIGNMENT-1

1. What is Linux Operating System (OS)? List three pros and cons of it.

- ❖ Linux Operating System (OS) is a free system software that makes your computer work.
- ❖ It connects the computer's hardware (like CPU, memory, and storage) with the applications you use.

Pros

- ❖ It is free software.
- ❖ It is safe from virus than windows.
- ❖ Fast and stable.

Cons

- ❖ Doesn't work for all software.
- ❖ Hard for beginners .
- ❖ Have hardware issues.

2.Differentiate between **Linux, Mac, Android**, and **Windows OS** with at least six unique features.

Features	Linux	Mac	Android	Windows
Cost	Free & open-source	Paid	Free	Paid
Performance	Very fast, efficient on low hardware	Smooth, optimized for Apple hardware	Varies (depends on phone brand)	Good, but needs stronger hardware
Updates	Frequent, community-driven	Controlled, less frequent but stable	Frequent (depends on manufacturer)	Regular updates
File System	ext4, XFS, Btrfs, etc.	APFS, HFS+	ext4 (Linux-based)	NTFS, FAT32, exFAT
User Friendliness	Hard for beginners	Very user-friendly	Simple, touch-friendly	Easy for beginners ,widely used
Security	Highly secure, few viruses	Very secure	Secure, but depends on updates	More prone to viruses & malware

3. Why is Linux preferred for **Mainframe Servers** for legacy application? Give three out-of-the-box technical reasons.

- ❖ Linux is preferred for Mainframe Servers for running legacy applications because it combines modern OS features with mainframe reliability.

The three technical reasons are:

- ❖ Efficient Resource Management.
- ❖ High Availability, Scalability & Fault Tolerance.
- ❖ Compatibility with Modern Networking Protocols.

4. Explain the structure of the Linux File System with proper diagram.

Note: you can use the **tree** command to find it out.

- ❖ / (Root) - The top-level directory, containing all other directories and files.
- ❖ /bin - Contains essential user command binaries.
- ❖ /home - Contains individual user home directories.

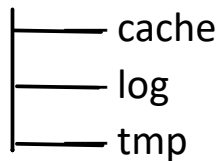
```
└─ user1
└─ user2
```

- ❖ /root - The home directory for the root user.
- ❖ /usr - A secondary hierarchy for user programs and data.

```
├─ bin
├─ lib
└─ local
    └─ bin
    └─ lib
    └─ src
```

- ❖ /etc - Holds system-wide configuration files.
- ❖ /lib - Stores essential shared libraries required by the binaries.

- ❖ /opt- Used for installing third party software.
- ❖ /dev – Contains device files, which represent hardware devices.
- ❖ /var - Contains variable data files, such as log files, mail queues .



5.If Linux OS is open-source, how do companies like Red Hat still making money from it? Do a market study and answer properly .

- ❖ Red Hat is a US-based company and a subsidiary of IBM that provides open-source enterprise software solutions.
- ❖ Support & Updates:
 - Linux is free, but companies want someone to fix the bugs in it.
 - Red Hat sells **support contracts** – like “tech support” with guaranteed help.
- ❖ Security & Reliability:
 - Red Hat gives safe, tested, and stable versions but, Big companies will pay as they don’t want server to crash.
 - It has maintains high security in them.
- ❖ Certifications:
 - Red Hat Linux is certified to work with many servers, apps, and hardware.
- ❖ Cloud & New Products:
 - Red Hat also sells tools like OpenShift.
 - Many companies pay for advanced tools.

So because of this even though it is open source it has high marketing business.

6. Write the command to display today's date and time (i.e., current-System time).

❖ bash
date

Sun Sep 21 22:39:42 IST 2025

7. Which **command** is used to check how long the system has been running.

❖ bash
uptime

8. What is the difference between shutdown -h now and halt?

shutdown -h	halt
Shutdown the system immediately.	Stops the function of CPU.
Notifies all logged-in users about shutdown.	Directly stops the system without any notifications.
Gracefully stops all processes.	May not gracefully stop all processes
Safe and recommended way to shut down.	Quick stop but not as safe as shutdown.
Shutdown immediately & halt.	Halt the system immediately.
Halts and then powers off.	Just halts CPU.

9. Compare init 0 and shutdown -h. Which is safer? Why?

Features	init 0	shutdown -h
Means	Switch to run-level 0 .	Shutdown and halt system safely.
Behaviour	Immediately stops all the processes.	Notifies before poweroff .
Safety	Less safety as it power-off immediately. Risk of data loss is high.	Safer as it notifies before shutdown. Ensures that the data is safely before closing.

Therefore shutdown -h is more likely to be safe compared to init 0.

10. A system administrator accidentally powers off a Server machine without shutting it down properly. What problems can occur to the said Server?

If the administrator power-off accidentally then the problem can occur at:

❖ Data loss:

Unsaved data in the memory or opened files can be lost.

❖ Loss of cache data:

Data stored temporarily in cache may never reach the disk.

❖ Increased Maintenance Effort:

System administrator need to spend extra time for running the recovery tools.

❖ Application error:

Application that may need proper shutdown fails to start.

❖ Boot Issues:

The server may fail to boot the system properly next time due to incomplete shutdown .

❖ Network issues:

Active network connections or sessions may terminate unexpectedly, causing client-side errors.

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